



Memory Diagram and Function-Writing Practice

Reminders

Re: Assignments:

- **EX09: Linked List Utility Functions** due at 11:59pm on LDOC (April 27)
 - Late submission deadline: 4pm on April 30th (start of the final exam)
 - You can work with a partner (or on your own)!!

Re: Final Exam:

- **4–7pm on Thursday, April 30**
 - Makeup final exam on Friday, May 1 in SN014
 - If you qualify for the makeup, please complete our [internal form](#) ASAP.
You will get a confirmation email next week!

[Apply to be a TA](#) for COMP110 in Fall 2026!

```
1 def select(  
2     input_data_table: dict[str, list[str]], selected_columns: list[str]  
3 ) -> dict[str, list[str]]:  
4     """Returns only the selected columns table into table."""  
5     result: dict[str, list[str]] = {}  
6  
7     for column_name in input_data_table:  
8         if column_name in selected_columns:  
9             result[column_name] = input_data_table[column_name]  
10  
11     return result  
12  
13 dt: dict[str, list[str]] = {"year": ["2018", "2019"], "temp": ["48.24", "46.  
63"], "result": ["no shadow", "shadow"]}  
14 cols: list[str] = ["temp", "result"]  
15  
16 dt_selected: dict[str, list[str]] = select(dt, cols)  
17 print(dt_selected)
```

```

1  def select(
2      input_data_table: dict[str, list[str]], selected_columns: list[str]
3  ) -> dict[str, list[str]]:
4      """Returns only the selected columns table into table."""
5      result: dict[str, list[str]] = {}
6
7      for column_name in input_data_table:
8          if column_name in selected_columns:
9              result[column_name] = input_data_table[column_name]
10
11         result["temp"].append("0")
12
13     return result
14
15     dt: dict[str, list[str]] = {"year": ["2018", "2019"], "temp": ["48.24", "46.
16     63"], "result": ["no shadow", "shadow"]}
17
18     cols: list[str] = ["temp", "result"]
19
20     dt_selected: dict[str, list[str]] = select(dt, cols)
21     print(dt_selected)

```

Consider the **new line** that was added to our function body. Discuss:

1. What would this line of code do?
2. Which variables/parameters would be mutated as a result of this line?

NOTE: DO NOT actually include this line in your select function in EX09!

Function-writing practice

Write a function called **shape** that:

- Has one parameter, **input_data_table**, of type `dict[str, list[str]]`
- It should return a `dict[str, int]` where each key is a column name from **input_data_table** and its corresponding value is the number of elements in that column.

Example behavior:

```
>>> data = { "name": ["Alice", "Bob", "Carol"], "major": ["CS", "Math"] }
```

```
>>> shape(data)
```

```
{"name": 3, "major": 2}
```